Exercise 2

1. Consider a portfolio of 300 shares of firm A worth \$10/share and 50 shares of firm B worth \$40/share. You expect a return of 8% for stock A and a return of 13% for stock B.

   (a) What is the total value of the portfolio, what are the portfolio weights and what is the expected return?

   (b) Suppose firm A's share price goes up to \$12 and firm B's share price falls to \$36. What is the new value of the portfolio? What return did it earn? After the price change, what are the new portfolio weights?

2. The Capital Asset Pricing Model

Suppose that the random return $\tilde{r}_M$ on the market portfolio has expected value $E(\tilde{r}_M) = 0.07$ and variance $\sigma_M^2 = 0.02$ and that the return on risk-free assets is $r_f = 0.02$.

a. According to the capital asset pricing model, if a risky asset has a random return $\tilde{r}_j$ with expected value $E(\tilde{r}_j) = 0.145$, what is the numerical value of this asset's beta β_j?

b. According to the capital asset pricing model, what is the numerical value of the covariance between this risky asset's random return $\tilde{r}_j$ and the random return $\tilde{r}_M$ on the market portfolio?

c. According to the capital asset pricing model, if another risky asset has a random return $\tilde{r}_k$ with expected value $E(\tilde{r}_k) = 0.01$, what is the numerical value of this asset's beta β_k?

3. Suppose that the random return $\tilde{r}_M$ on the market portfolio has expected value $\mathrm{E}(\tilde{r}_M) = 0.07$ and variance $\sigma_M^2 = 0.02$ and that the return on risk-free asset is $r_f = 0.02$. (Question from previous exam)

   (a) According to CAPM, if risky asset j has a random return $\tilde{r}_j$ with expected value $\mathrm{E}(\tilde{r}_j) = 0.145$, what is the numerical value of this asset's beta β_j?

   (b) What is the numerical value of the covariance between this risky asset's random return $\tilde{r}_j$ and the random return $\tilde{r}_M$ on market portfolio?

   (c) If another risky asset has a random return $\tilde{r}_k$ with expected value $\mathrm{E}(\tilde{r}_k) = 0.01$, what is the numerical value of this asset's beta β_k?